

# CloudFront Signed URL Process

- Two types of signers:
  - Either a trusted key group (recommended)
    - Can leverage APIs to create and rotate keys (and IAM for API security)
  - An AWS Account that contains a CloudFront Key Pair
    - Need to manage keys using **the root account and the AWS console**
    - Not recommended because you shouldn't use the root account for this
- In your CloudFront distribution, create one or more **trusted key groups**
- You generate your own public / private key
  - The private key is used by your applications (e.g. EC2) to sign URLs
  - The public key (uploaded) is used by CloudFront to verify URLs

# CloudFront - Pricing

- CloudFront Edge locations are all around the world
- The cost of data out per edge location varies

| Per Month  | United States,<br>Mexico, &<br>Canada | Europe & Israel | South Africa,<br>Kenya, &<br>Middle East | South America | Japan   | Australia & New<br>Zealand | Hong Kong,<br>Philippines,<br>Singapore, South<br>Korea, Taiwan, &<br>Thailand | India   |
|------------|---------------------------------------|-----------------|------------------------------------------|---------------|---------|----------------------------|--------------------------------------------------------------------------------|---------|
| First 10TB | \$0.085                               | \$0.085         | \$0.110                                  | \$0.110       | \$0.114 | \$0.114                    | \$0.140                                                                        | \$0.170 |
| Next 40TB  | \$0.080                               | \$0.080         | \$0.105                                  | \$0.105       | \$0.089 | \$0.098                    | \$0.135                                                                        | \$0.130 |
| Next 100TB | \$0.060                               | \$0.060         | \$0.090                                  | \$0.090       | \$0.086 | \$0.094                    | \$0.120                                                                        | \$0.110 |
| Next 350TB | \$0.040                               | \$0.040         | \$0.080                                  | \$0.080       | \$0.084 | \$0.092                    | \$0.100                                                                        | \$0.100 |
| Next 524TB | \$0.030                               | \$0.030         | \$0.060                                  | \$0.060       | \$0.080 | \$0.090                    | \$0.080                                                                        | \$0.100 |
| Next 4PB   | \$0.025                               | \$0.025         | \$0.050                                  | \$0.050       | \$0.070 | \$0.085                    | \$0.070                                                                        | \$0.100 |
| Over 5PB   | \$0.020                               | \$0.020         | \$0.040                                  | \$0.040       | \$0.060 | \$0.080                    | \$0.060                                                                        | \$0.100 |

lower

higher

# CloudFront – Price Classes

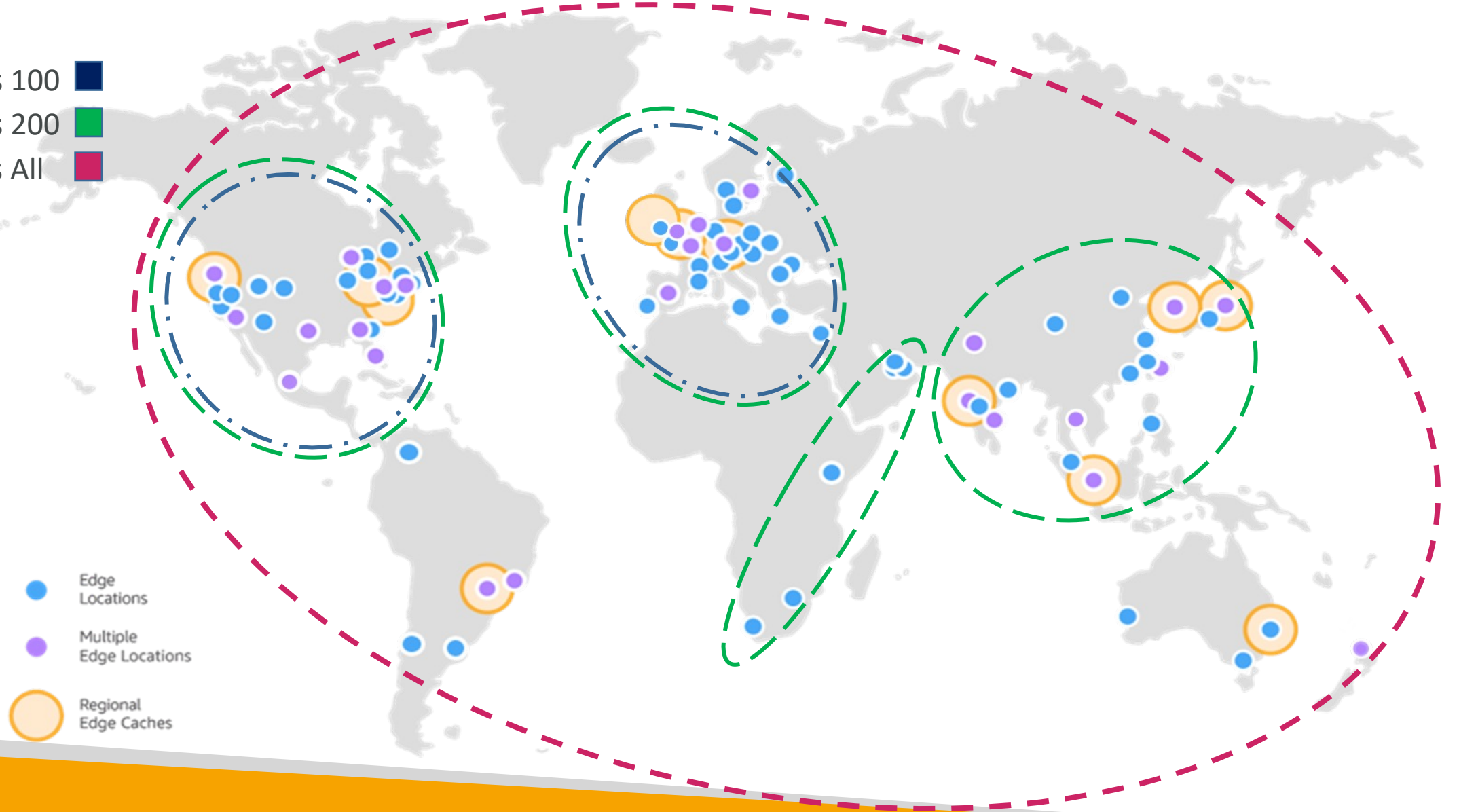
- You can reduce the number of edge locations for **cost reduction**
- Three price classes:
  1. Price Class All: all regions – best performance
  2. Price Class 200: most regions, but excludes the most expensive regions
  3. Price Class 100: only the least expensive regions

| Edge Locations Included Within | United States, Mexico, & Canada | Europe & Israel | South Africa, Kenya, & Middle East | South America | Japan | Australia & New Zealand | Hong Kong, Philippines, Singapore, South Korea, Taiwan, & Thailand | India |
|--------------------------------|---------------------------------|-----------------|------------------------------------|---------------|-------|-------------------------|--------------------------------------------------------------------|-------|
| Price Class All                | Yes                             | Yes             | Yes                                | Yes           | Yes   | Yes                     | Yes                                                                | Yes   |
| Price Class 200                | Yes                             | Yes             | Yes                                | x             | Yes   | x                       | Yes                                                                | Yes   |
| Price Class 100                | Yes                             | Yes             | x                                  | x             | x     | x                       | x                                                                  | x     |

# CloudFront - Price Class

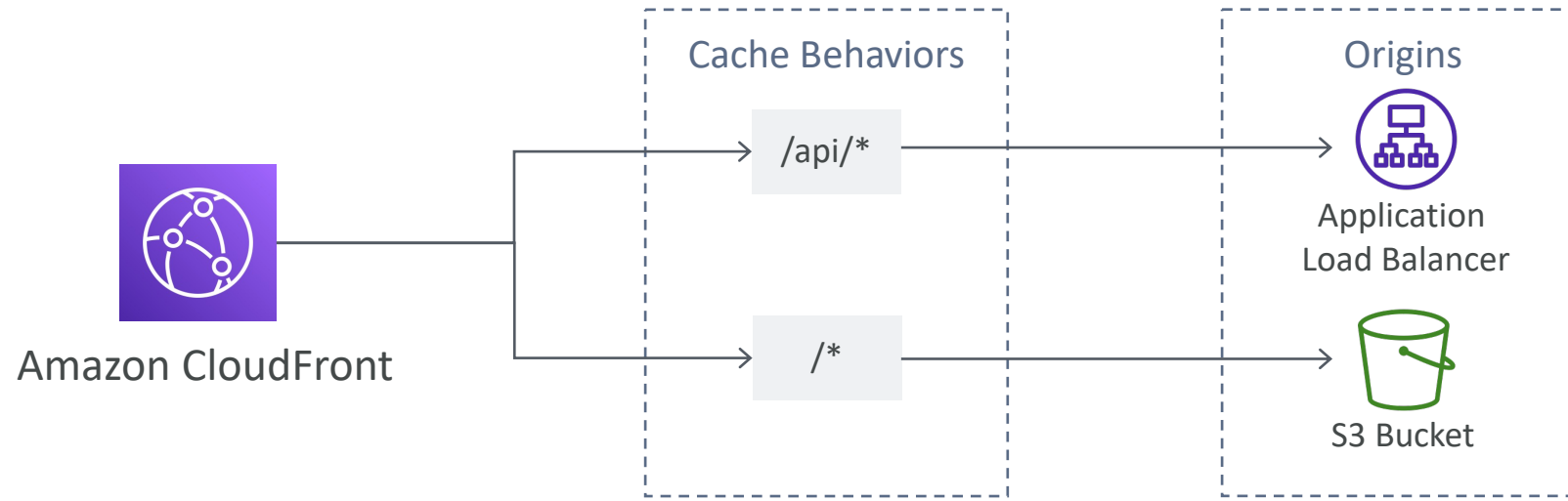
Prices Class 100 ■  
Prices Class 200 ■  
Prices Class All ■

● Edge Locations  
● Multiple Edge Locations  
○ Regional Edge Caches



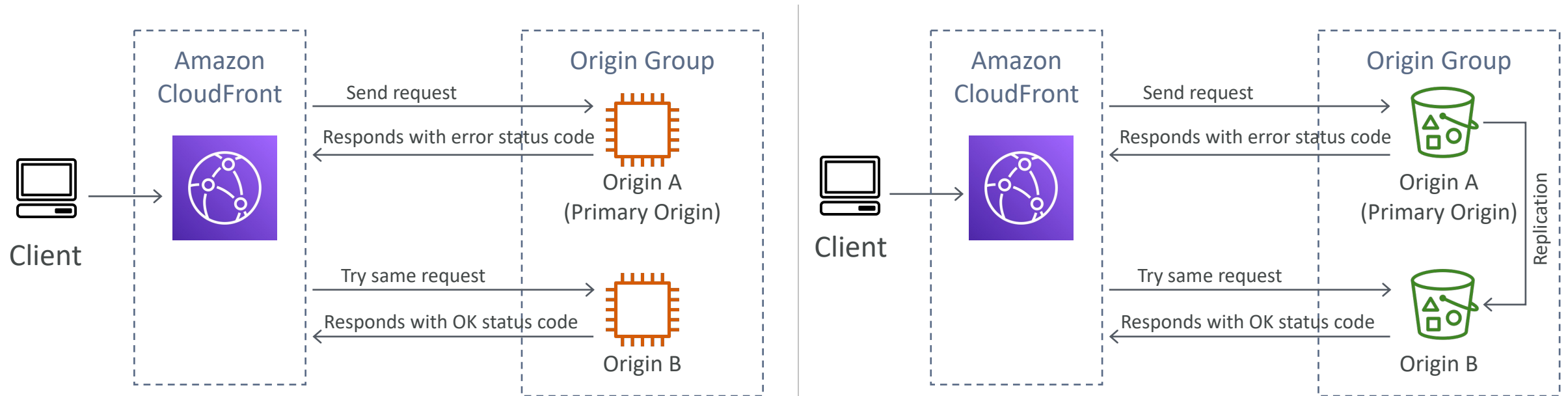
# CloudFront – Multiple Origin

- To route to different kind of origins based on the content type
- Based on path pattern:
  - /images/\*
  - /api/\*
  - /\*



# CloudFront – Origin Groups

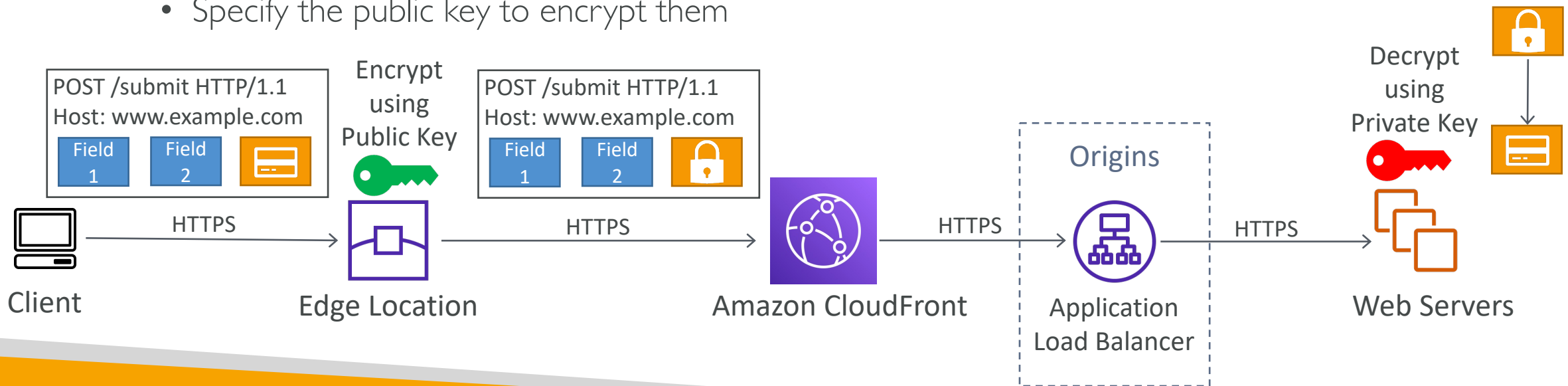
- To increase high-availability and do failover
- Origin Group: one primary and one secondary origin
- If the primary origin fails, the second one is used



S3 + CloudFront – Region-level High Availability

# CloudFront – Field Level Encryption

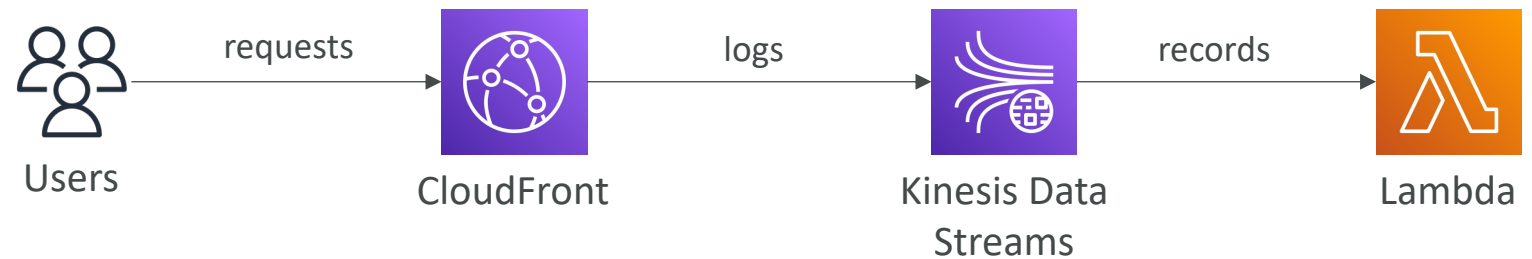
- Protect user sensitive information through application stack
- Adds an additional layer of security along with HTTPS
- Sensitive information encrypted at the edge close to user
- Uses asymmetric encryption
- Usage:
  - Specify set of fields in POST requests that you want to be encrypted (up to 10 fields)
  - Specify the public key to encrypt them



# CloudFront – Real Time Logs

- Get real-time requests received by CloudFront sent to Kinesis Data Streams
- Monitor, analyze, and take actions based on content delivery performance
- Allows you to choose:
  - **Sampling Rate** – percentage of requests for which you want to receive
  - Specific fields and specific Cache Behaviors (path patterns)

## Real-time Processing



## Near Real-time Processing

